

NEEDLES, CA, USA

WMO: 723805

Lat: 34.768N

Lon: 114.619W

Elev: 271

StdP: 98.11

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 23179

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	3.0	4.6	-18.8	0.7	18.5	-16.5	0.9	19.1	13.5	14.5	12.1	14.7	2.4	240	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	46.0	20.9	44.6	20.7	43.5	20.5	24.9	37.4	24.1	36.9	23.5	36.9	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.9	30.4	20.2	15.4	31.5	18.8	14.0	33.0	76.9	37.2	73.7	36.6	71.2	36.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.6	10.1	8.6	DB	0.0	48.4	1.8	1.2	-1.3	49.2	-2.4	49.9	-3.4	50.5	-4.7	51.4
			WB	-3.9	25.6	1.6	1.2	-5.0	26.5	-5.9	27.2	-6.8	27.8	-7.9	28.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.3	13.2	15.0	19.2	23.0	28.0	34.0	36.8	35.8	31.9	24.5	17.8	12.2
	DBStd	9.22	3.22	3.51	3.87	3.83	4.03	3.11	2.49	2.48	3.22	3.62	3.78	3.15
	HDD10.0	29	11	3	0	0	0	0	0	0	0	0	1	14
	HDD18.3	554	162	102	37	7	0	0	0	0	0	3	54	189
	CDD10.0	5262	108	142	286	391	559	720	831	799	657	451	234	83
	CDD18.3	2746	1	8	64	148	301	470	573	541	407	196	37	1
	CDH23.3	44313	12	104	752	1858	4533	8292	10386	9494	6367	2190	319	5
CDH26.7	30319	1	15	277	896	2781	5996	7915	7048	4247	1062	82	0	
Wind	WSAvg	3.6	3.6	3.5	3.7	4.1	4.0	3.9	3.6	3.4	3.2	3.5	3.4	3.4
Precipitation	PrecAvg	117	15	16	13	6	2	1	7	15	13	8	10	11
	PrecMax	243	92	91	81	71	19	12	56	86	120	42	44	66
	PrecMin	23	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	60	22	20	17	14	4	2	12	22	22	10	11	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	28.5	34.5	38.3	42.9	47.9	48.3	46.4	43.8	38.0	31.4	24.0
		MCWB	11.4	12.8	15.4	16.1	19.4	19.5	21.7	21.3	20.4	18.4	15.0	12.8
	2%	DB	22.7	26.5	31.8	35.9	41.0	45.2	46.4	45.1	42.2	35.9	29.0	21.7
		MCWB	10.6	12.1	14.2	15.5	17.8	18.8	21.5	21.5	20.1	17.3	13.9	11.0
	5%	DB	21.0	24.3	29.7	33.8	39.0	43.8	45.0	43.8	41.0	34.1	27.1	19.8
		MCWB	10.0	11.2	13.5	14.7	17.1	18.8	21.5	21.3	20.0	16.6	13.4	10.1
	10%	DB	19.3	22.2	27.8	31.7	37.2	42.4	43.5	42.6	39.6	32.4	24.9	18.0
		MCWB	9.4	10.6	12.6	14.0	16.5	18.6	21.2	21.2	19.7	16.1	12.4	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.1	16.6	18.3	21.6	22.9	25.9	26.3	24.5	21.0	16.8	14.2
		MCDB	19.1	22.5	30.9	33.1	40.0	41.1	41.7	38.8	35.4	30.3	26.6	19.8
	2%	WB	13.0	13.7	15.1	16.4	19.2	21.6	24.7	25.2	23.3	19.6	15.2	13.0
		MCDB	18.0	20.9	28.5	32.1	37.5	40.9	37.8	37.0	35.1	30.6	26.4	18.8
	5%	WB	11.8	12.5	14.1	15.5	17.9	20.5	24.0	24.3	22.5	18.5	14.1	11.7
		MCDB	18.1	20.8	27.7	30.9	36.6	40.3	37.2	36.8	34.9	30.9	24.4	17.6
	10%	WB	10.6	11.6	13.2	14.6	16.9	19.6	23.5	23.7	21.7	17.3	13.1	10.4
		MCDB	17.2	20.4	26.0	29.8	35.5	39.8	37.4	36.7	34.8	29.7	23.4	16.4
Mean Daily Temperature Range	5% DB	MDBR	10.9	12.0	13.6	13.7	14.0	14.4	12.5	12.7	13.5	13.1	12.1	10.3
		MCDBR	12.8	15.1	16.5	16.4	16.4	15.9	14.0	13.9	14.8	15.0	14.1	12.1
		MCWBR	6.4	7.0	7.0	6.9	6.5	5.6	5.0	4.9	5.7	6.7	6.6	6.0
	5% WB	MCDBR	10.7	12.5	15.0	14.6	15.3	14.3	11.6	11.8	12.2	12.6	12.5	10.2
		MCWBR	5.6	6.0	6.5	6.4	7.0	6.4	4.2	4.2	5.1	6.0	6.1	5.4
Clear-Sky Solar Irradiance	taub		0.273	0.276	0.290	0.324	0.335	0.334	0.384	0.379	0.338	0.301	0.277	0.272
	taud		2.598	2.612	2.546	2.422	2.419	2.424	2.394	2.401	2.484	2.564	2.606	2.615
	Ebn at Noon		952	984	989	961	948	943	894	895	922	939	939	933
	Edh at Noon		75	83	96	115	117	116	119	116	101	86	74	70
All-Sky Solar Radiation	RadAvg		3.27	4.26	5.74	7.12	8.04	8.54	7.59	7.05	6.24	4.83	3.63	2.92
	RadStd		0.21	0.32	0.25	0.25	0.26	0.28	0.34	0.29	0.23	0.23	0.15	0.20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-2.64	N/A	-1.00	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	+0.39	N/A	-1.70	N/A	-0.41	N/A	N/A	N/A	+143	+80

Nomenclature: See separate page